

CyberGraph

Explainable AI-Powered Network Intrusion Detection System
with Behavioral Attack-Route Intelligence & Real-Time SOC Dashboard

Project Brief | Dataset: NF-UNSW-NB15-v3 | Generated: 06 September 2026

■ Project Overview

CyberGraph is a **real-time, multi-model AI/ML Network Intrusion Detection System (NIDS)** that captures live network packets via Scapy, extracts 49 NetFlow features per flow, classifies traffic using a stacked ensemble of supervised (XGBoost) and unsupervised (Autoencoder) models, further refines risk through a custom Dynamic Bipartite Temporal GNN, and presents all detections through a live Security Operations Center (SOC) dashboard. The system introduces two novel contributions: a **Dynamic Bipartite Temporal GNN** with GRU-based node memory, and a **Behavioral Attack-Route Intelligence (BARI)** layer that detects coordinated, multi-stage attacks earlier than conventional flow-level NIDS.

- **Dataset:** NF-UNSW-NB15-v3 — 377,279 labeled flows, 55 features, 9 attack categories
- **End-to-end detection latency:** 3.81 ms per flow (live measured on Wi-Fi traffic)
- **XGBoost test-set accuracy:** 99.99% on held-out NF-UNSW-NB15-v3 test split
- **Live capture:** Scapy + WinPcap/Npcap — real-time physical packet ingestion
- **BARI baseline:** learned from 198,784 benign training flows (single-pass, O(N))

■ 9-Stage Detection Pipeline

Stage	Component	What It Does
1	Scapy Packet Capture	Live capture from Wi-Fi / NIC; capture latency measured in ms
2	Feature Extraction	49 NetFlow features/flow (bytes, packets, duration, IAT, TCP flags, etc.)
3	XGBoost Classifier	Supervised binary classifier → P(attack); trained on NF-UNSW-NB15-v3
4	Autoencoder Anomaly	Unsupervised; AE score = $MSE / ae_mse_max \in [0,1]$ — no $\times 10$ scaling
5	Fusion Score	Logistic meta-learner fuses XGBoost + Autoencoder into single risk score
6	Dynamic Temporal GNN	HeteroBipartiteConv + GRU memory; scores HOST↔SERVICE edges over time
7	Risk Classification	Assigns Level 1/2/3 + attack type (DoS, Recon, Exploits, Backdoor, Fuzzer)
7b	BARI Layer	Route Novelty + Surface Progression fused with GNN risk (novel contribution)
8	Correlation Engine	Rule-based temporal tracking; detects kill-chain stage per attacker IP
9	Response Engine	Recommends / executes OS-level firewall block (Windows netsh rule)

■ AI / ML Architecture

a) Dynamic Bipartite Temporal GNN

- Strict bipartite graph $G = (V_{\text{host}}, V_{\text{service}}, E)$ — no host-to-host edges, bipartite validated at runtime
- **HeteroBipartiteConv**: aggregates structural context between Host nodes and Service nodes
- **TemporalNodeUpdate (GRU)**: preserves entity behavioral memory across snapshots: $h_{t-1} \rightarrow h_t$
- **EdgeRiskClassifier (MLP)**: 3-layer MLP with BatchNorm + Dropout; classifies each edge as attack/benign
- 500-flow sliding window; GRU state persists across all inference calls (online learning)
- 8-dimensional leakage-free node features — zero ground-truth labels in feature vectors

b) BARI — Behavioral Attack-Route Intelligence (Novel Contribution)

Signal	Formula / Definition
Route Novelty (RN)	$RN = 1 - P(\text{service} \mid \text{benign_baseline})$, range [0, 1]
Attack-Surface Progression (ASP)	ASP = max sensitivity score of recent dst_ports accessed by src IP
BARI Fusion Score	$BARI = 0.30 \times RN + 0.30 \times ASP + 0.40 \times GNN_risk$
Sensitivity examples	Port 80 = 0.10 SSH:22 = 0.65 RDP:3389 = 0.85 Backdoor:4444 = 1.00
Proven escalation	Port 80 → BARI 0.26 SSH → 0.74 Backdoor port → 0.96

BARI is a rule-and-statistics layer — it does not contain a trainable neural network. Alpha=0.30, Beta=0.30 are documented hyperparameters; GNN_risk weight = 0.40.

c) Explainable AI (SHAP)

- SHAP TreeExplainer on XGBoost — per-flow feature saliency Z-scores
- ExplainableGraphAI: why_this_attack_reasons list generated for every Level 2+ alert
- Primary saliency driver displayed on each threat card in the SOC dashboard

■ Temporal Correlation & Kill-Chain Detection

The **CorrelationEngine** is a rule-and-statistics layer (not an ML model) that tracks per-source-IP event histories in a rolling deque (max 50 events) and detects escalating attack patterns using exponential temporal decay scoring.

Property	Detail
Correlation Score	$C = (1/N) \times \sum r_i \times 0.85^{(N-1-i)}$ — exponential temporal decay; recent events dominate
Observable Kill-Chain Stages	RECONNAISSANCE → DELIVERY → EXPLOITATION → COMMAND_AND_CONTROL → A
Escalation Detection	Strictly increasing GNN risk across last 4 consecutive events
Max Events Per IP	50 events (rolling deque — oldest evicted automatically)
Correlation Threshold	$C \geq 0.40$ required to appear in Graph Intelligence panel
Stage Omitted	Weaponization — occurs before delivery; not observable from NetFlow data

■ SOC Dashboard Features

Feature	Description
Live Threat Feed	Per-alert: risk score, attack type, geo flag, BARI score, SHAP explanation

Feature	Description
Attack Type Histogram	Live Canvas bar chart — DoS / Recon / Exploit / Backdoor / Fuzzer counts
Bipartite Graph Radar	Animated HOST↔SERVICE canvas with GNN-scored edges + particle flow
Graph Correlation Panel	Kill-chain timeline per attacker IP — stage badge + escalation alert
Explain Last Detection	Full 9-step pipeline modal showing actual measured values per stage
IP Geolocation	Flag + country + city on every alert (ip-api.com, in-memory cached)
Browser Notifications	Native OS notification + red banner triggered on risk ≥ 0.85
CSV Export	Rolling 500-incident log download (timestamp, IP, port, scores, geo, action)
PDF Incident Report	Print-ready HTML report: summary, attack breakdown, model accuracy tables
Dark / Light Theme	Instant theme toggle persisted via localStorage

■ Performance Metrics

Metric	Value	Notes
End-to-end latency	3.81 ms	Live measured on Wi-Fi capture
Packet capture latency	1.25 ms	Scapy WinPcap — physically measured
Feature extraction latency	0.64 ms	49-feature prep + StandardScaler transform
XGBoost accuracy	99.99%	NF-UNSW-NB15-v3 held-out test split
Autoencoder AE score	MSE / ae_mse_max	Normalised to [0,1]; no arbitrary multiplier
GNN sliding window	500 flows	GRU temporal state persisted across calls
BARI baseline	198,784 flows	Benign training flows (Label = 0)
Correlation engine	< 1 ms overhead	Pure Python deque + arithmetic

■ Technology Stack

Layer	Libraries / Tools
ML / Deep Learning	PyTorch, PyTorch Geometric (HeteroData), XGBoost, scikit-learn, SHAP
Network Capture	Scapy, WinPcap / Npcap
Graph Processing	NetworkX (BipartiteGraphBuilder), PyTorch Geometric
Backend / API	Python 3.14, Flask, joblib
Data Processing	Pandas, NumPy
Frontend	HTML5, CSS3 (CSS Variables / dark+light theme), Vanilla JS, Canvas API
Dataset	NF-UNSW-NB15-v3 — 377,279 flows, 55 features, 9 attack categories

■ Novel Research Contributions

- **BARI (Behavioral Attack-Route Intelligence):** Novel fusion of empirically-learned Route Novelty and Attack-Surface Progression with GNN edge risk — detects coordinated multi-stage attacks before kill-chain completion.
- **Dynamic Bipartite Temporal GNN:** Custom 3-layer architecture (HeteroBipartiteConv + GRU + EdgeRiskClassifier MLP) with GRU-based node memory on a strict HOST↔SERVICE bipartite topology — leakage-free node features validated at runtime.
- **Temporal CorrelationEngine:** Lightweight rule-based layer providing observable kill-chain stage detection with exponential decay scoring — end-to-end latency under 4 ms.
- **Unified event record:** Single event object powers threat feed, kill-chain timeline, BARI panel, pipeline explain, CSV export, and PDF report — no duplicate data sources.
- **Fully transparent pipeline:** Every detection stage's actual measured value exposed via /api/pipeline_explain — directly auditable and reproducible for IEEE peer review.

CyberGraph — Explainable AI NIDS with BARI | NF-UNSW-NB15-v3 | Generated 06 Sep 2026, 14:47